

DESIGN AND DEVELOPMENT OF A PAPER-BASED COLORIMETRIC IRON DETECTION KIT

ABSTRACT

This project focuses on the design and development of a low-cost, portable, and user-friendly paper-based colorimetric detection kit for iron (Fe^{3+}) in aqueous samples. The detection principle is based on coordination chemistry, where potassium thiocyanate (KSCN) reacts with ferric ions to form a blood-red colored complex. By integrating this well-established laboratory reaction into a paper-based analytical device (PAD), the project transforms a conventional chemical assay into a compact “lab-on-a-chip” format. The methodology follows the Innovation and Design Thinking (IDT) framework, emphasizing human-centered design, affordability, and accessibility. The developed prototype demonstrates visible color changes corresponding to iron concentration, making it suitable for use in resource-limited and remote areas without the need for advanced instrumentation.

CHAPTER 1: INTRODUCTION

1.1 Background

Iron is an essential micronutrient that plays a vital role in environmental systems, agriculture, and human health. However, excessive iron concentration in drinking water and soil can lead to adverse effects such as metallic taste, staining, pipe corrosion, and potential health concerns. Conventional iron detection techniques, including spectrophotometry and atomic absorption spectroscopy, require costly equipment, trained personnel, and laboratory infrastructure. These limitations restrict routine monitoring, especially in rural and under-resourced regions.

To address this gap, the present project repurposes the classical thiocyanate test for iron into a paper-based sensing platform. Paper-based analytical devices have gained significant attention due to their low cost, portability, biodegradability, and ease of use. Integrating chemical sensing with paper substrates aligns with modern interdisciplinary approaches combining chemistry, materials science, and design thinking.

1.2 Problem Statement

Existing iron detection methods are not suitable for frequent, on-site testing in remote or rural areas due to high costs, lack of portability, and dependence on laboratory facilities. There is a clear need for a human-centered, affordable, and simple iron detection system that provides quick and visually interpretable results without specialized training.

1.3 Objectives

The primary objectives of this project are:

- To design and develop a paper-based sensing platform for iron detection.
- To utilize potassium thiocyanate as a selective reagent for Fe^{3+} detection.
- To generate a visible colorimetric response that correlates with iron concentration.
- To apply the Design Thinking methodology to create an accessible and scalable diagnostic solution.

CHAPTER 2: LITERATURE SURVEY

2.1 Paper-Based Colorimetric Sensors

Recent advancements in Innovation and Design Thinking have encouraged the development of point-of-care (POC) diagnostic tools. Paper-based analytical devices (PADs) are increasingly recognized as an effective alternative to conventional sensors due to their low manufacturing cost, minimal reagent consumption, and ease of disposal. PADs have been widely explored for applications in water quality monitoring, medical diagnostics, and environmental analysis. Their ability to provide rapid, instrument-free detection makes them particularly suitable for low-resource settings.

2.2 Chemical Principle of Iron Detection

The detection of ferric ions using thiocyanate is a well-known qualitative and quantitative chemical test. When Fe^{3+} ions react with thiocyanate ions (SCN^-), a penta-aqua thiocyanatoiron(III) complex is formed, which exhibits a characteristic blood-red color. The intensity of this color is directly proportional to the concentration of Fe^{3+} ions present in the sample. This strong visual response makes the reaction ideal for colorimetric sensing applications.

CHAPTER 3: METHODOLOGY (DESIGN THINKING APPROACH)

The project follows the five stages of the Design Thinking framework: Empathize, Define, Ideate, Prototype, and Test.

3.1 Empathize and Define

The initial stage involved understanding the challenges faced by users in remote and rural regions, such as lack of laboratory access and limited technical expertise. The problem was defined as the need for a simple, portable, and affordable iron detection method that provides rapid results at the point of use.

3.2 Ideation

Based on the identified problem, the idea was to convert a traditional liquid-based chemical assay into a solid-state, paper-based format. Potassium thiocyanate was selected due to its strong and selective colorimetric response with Fe^{3+} ions. Paper was chosen as the substrate owing to its capillary action, availability, and eco-friendly nature.

3.3 Prototype Design and Fabrication

The prototype iron detection kit was developed using the following steps:

- Selection of suitable paper substrate such as Whatman filter paper.
- Creation of defined reaction zones on the paper.
- Integration of potassium thiocyanate chips or impregnated zones onto the paper.
- Drying and stabilization of the reagent on the paper matrix.
- Application of aqueous iron samples of varying concentrations onto the sensing zones.

A conceptual design sketch of the prototype includes a small paper strip with marked test zones where the sample is added, resulting in visible color formation.

CHAPTER 4: MATERIALS AND PRINCIPLE

4.1 Materials Used

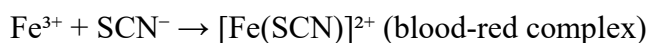
- Whatman filter paper or equivalent paper substrate
- Potassium thiocyanate (KSCN)
- Standard ferric ion (Fe^{3+}) solutions of known concentrations



- Distilled water
- Dropper or micropipette
- Protective gloves and laboratory glassware

4.2 Principle of Operation

The working principle of the kit is based on coordination chemistry. When an aqueous sample containing Fe^{3+} ions comes into contact with the potassium thiocyanate impregnated paper, a coordination complex is formed according to the reaction:



The intensity of the red color developed on the paper is directly related to the concentration of Fe^{3+} ions present in the sample, allowing semi-quantitative analysis through visual comparison.

CHAPTER 5: PROCEDURE

1. Prepare the paper-based sensor by impregnating defined zones with potassium thiocyanate solution and allowing it to dry.
2. Place the paper strip on a clean, flat surface.
3. Add a few drops of the test water sample onto the reagent zone.
4. Observe the color change within a few seconds.
5. Compare the observed color with a pre-prepared color intensity scale to estimate iron concentration.



CHAPTER 6: RESULTS AND DISCUSSION

6.1 Colorimetric Response

The developed kit exhibited a clear and distinct color change upon exposure to ferric ions. Samples with low iron concentration produced a pale pink or orange coloration, while higher concentrations resulted in a deep blood-red color. This gradient response confirms the sensitivity of the paper-based sensor.



Strip colour indicating presence of iron



Colour gradient seen in iron samples

6.2 Limitations and Innovations

Currently, the kit selectively detects Fe^{3+} ions and does not account for Fe^{2+} ions present in the sample. However, this limitation also presents an opportunity for innovation. By incorporating an oxidizing agent such as hydrogen peroxide (H_2O_2), Fe^{2+} can be converted to Fe^{3+} , enabling total iron estimation.

CHAPTER 7: FUTURE SCOPE

- Integration of a smartphone-based mobile application to analyze color intensity and provide quantitative results.
- Expansion of the kit to detect total iron by incorporating oxidation chemistry.
- Large-scale production and packaging for community-level water testing programs.
- Extension of the paper-based platform for detection of other heavy metals and contaminants.

CHAPTER 8: SUSTAINABLE DEVELOPMENT GOALS (SDG) RELEVANCE

This project aligns with several United Nations Sustainable Development Goals:

- **SDG 3: Good Health and Well-being** – Ensures safer drinking water by enabling iron monitoring.



- **SDG 6: Clean Water and Sanitation** – Promotes affordable and accessible water quality testing.



CONCLUSION

The paper-based colorimetric iron detection kit successfully demonstrates how interdisciplinary thinking and design-driven innovation can transform a conventional laboratory reaction into an accessible, real-world solution. By combining chemistry, material science, and human-centered design, the project offers a sustainable and scalable approach to iron detection, particularly beneficial for resource-limited.